

	IA-427 Forensics Report Internal Case Mark Jones, <i>Animals Today</i> (AT) Incident Status: In progress	Report written by: Dickson Mogaka Revision: 1
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IA 427 instructions:

- Download the Mark Jones image from our share
 - Import into Autopsy
 - File a report using the template below
- =====

1. Executive Summary

Mark Jones, a senior writer at *Animals Today* (AT), is under investigation for suspected involvement in the illegal harvesting, sale, and distribution of elephant tusks. Although his position requires frequent travel to various regions of Africa to report on wildlife conservation issues, investigators believe Mark may have used these trips as an opportunity to engage in black-market ivory trafficking. Allegations indicate that during his assignments abroad, he obtained raw tusks and coordinated their sale through illicit channels.

Beyond the suspected wildlife crimes, Mark is also accused of misusing AT's corporate resources to support his activities. This includes the possible use of company-issued devices, communication tools, and travel privileges to facilitate the transport, documentation, or sale of contraband materials. AT leadership is concerned that Mark may have exploited his professional access to restricted wildlife areas and conservation contacts to further his illegal operations.

AT policy strictly prohibits the use of removable storage devices unless explicitly authorized by leadership. In approved situations, AT provides a company-owned storage device for employee use. Mark was issued a corporate USB drive for his trips to Africa.

Mark is well known among colleagues as a sports car enthusiast and frequently stores or shares high-resolution images of exotic vehicles. While this interest is not inherently suspicious, investigators have noted that such images routinely appear on his devices, including the corporate USB drive assigned for international travel. This USB drive—issued specifically for his reporting assignments in Africa—has become a central piece of digital evidence. AT's security team has requested a comprehensive forensic examination of the device to determine whether it contains materials related to ivory trafficking, evidence of policy violations, or indications of personal misuse of company property.

Nature of the case: The investigation aims to determine whether Mark has been involved in tusk harvesting to support illegal wildlife trade, whether any digital artifacts link him to the acquisition or sale of elephant tusks, and whether his actions constitute violations of AT's corporate policies and ethical standards.

Evidence source- USB storage device. Chain of custody was established.

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2. The information security policies in the company include the following:

1. **Confidentiality of Company Property:** All confidential paper documents and electronic files belong exclusively to ABC and must not be distributed outside the organization.

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2. **Removable Media Prohibition:** The use of removable storage devices is strictly prohibited unless explicitly authorized by AT leadership. In approved cases, AT will issue a company-owned storage device to the employee for use.
3. **Data Transfer:** Employees may not transfer data from AT to other devices or accounts. When a mass transfer of such data is required, we ask employees to contact our [Security Specialists] for help.
4. **Personal Data Restrictions:** Employees are prohibited from storing personal data on ABC-owned computers.
5. **Acceptable Use of Computing Resources:** All ABC electronic and computing resources are company property and may only be used for legitimate business purposes. ABC reserves the right to inspect and investigate its equipment at any time and for any reason.

Things to be aware of for your investigation:

1. Any image, document, or spreadsheet that discusses tusks will be evidence of wrongdoing.
2. There are 13 artifacts in this case.
3. An easter egg(s) **may** also be found

3. Drive image hash

Drive/Image Verify Results	
Sector count	30031872
MD5 Hash	
Computed hash	cc812d52cd627d30885bc37b15d1a525
Report Hash	cc812d52cd627d30885bc37b15d1a525
Verify result	Match
SHA1 Hash	
Computed hash	34d53db1bfe8f2b966d49aea3189169fc710e
Report Hash	34d53db1bfe8f2b966d49aea3189169fc710e
Verify result	Match
Bad Blocks List	
Bad block(s) in image	No bad blocks found in image

4. Tool Listing

1. Autopsy 4.21.0 Autopsy is an open-source digital forensics toolkit for processing images of hard drives and removable USB devices. It has many plugins to assist in data analysis and review. See <http://www.sleuthkit.org/autopsy/> for additional information or a copy of the software.
2. WinMD5 v1.2.0 WinMD5 is a tool that takes file evidence as input and produces the MD5 hash as output. It can also compare a second copy of the file to confirm hash numbers. Visit [Wind6.com](http://www.winmd5.com/) for more details
3. [Hexed.it](http://hexed.it/) is an online hex editor that lets you view and edit the raw hexadecimal data of a file directly in your browser.

5. Artifact Analysis –

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1	Artifact number:	1
2	Name of file:	car4.jpg
3	Hash:	94f7419f2b296981c0e0d9e302a5af89
4	File description: Provide the zone ID and URL	Zoneld=3 ReferrerUrl=https://www.google.com/ HostUrl=https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcTR8P58RFYoUEpdiXCB3XM_C_BGkDio_T2ppQ&s
5	Location/discovery method:	Located on Mark Jones USB drive, recovered via Autopsy.
6	Work done to recover:	Standard bit-stream copy. Extracted from Autopsy
7	Notice of suspicious work:	The photo/file had hidden in it another file. Seeming to avoid detecting and be a secret.
8	Direction of evidence/ Attribution:	This file is an image of a car that has a photo of tusk hidden in it and supports the charge. Attribution will be determined with further investigation
9	Policy Analysis:	Acceptable Use of Computing Resources: This file was used to transport data about tusks in secret in a possibly illegal way Personal Data Restrictions- Mark Jones has a personal file about cars on company property Removable Media Prohibition– This file was found on a USB drive plugged into Dustin Green’s computer.



1	Artifact number:	2
2	Name of file:	car2.jpg
3	Hash:	51e9361127aca7965bace981bfd8c022
4	File description: Provide the zone ID and URL	Zoneld=3 ReferrerUrl=https://www.google.com/ HostUrl=https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQj5Y8P9UC2eNOEayhfNSoNAOur1CV01XVgVg&s

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5	Location/discovery method:	Located on Mark Jones USB drive, recovered via Autopsy.
6	Work done to recover:	Standard bit-stream copy. Extracted from Autopsy
7	Notice of suspicious work:	The photo/file had hidden in it another file. Seeming to avoid detecting and be a secret.
8	Direction of evidence/ Attribution:	This file is an image of a car that has a photo of tusk hidden in it and supports the charge. Attribution will be determined with further investigation
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1	Artifact number:	3
2	Name of file:	car3.jpg
3	Hash:	a6eb8de1153c378c209c7ddc1722601c
4	File description: Provide the zone ID and URL	Zoneld=3 ReferrerUrl=https://www.google.com/ HostUrl=https://encrypted-tbn0.gstatic.com/images?q=tbn:AND9GcRaqcCQU977qa7jT-IOgHS80tqwZcMS7ZsutA&s
5	Location/discovery method:	Located on Mark Jones USB drive, recovered via Autopsy. Stenography was used. Revealed using Hexed.it
6	Work done to recover:	Use Hexed.it and identified each photo by using magic numbers to identify the beginning and end of a photo
7	Notice of suspicious work:	The photo/file was hidden in another file. Seeming to avoid detecting and be a secret.
8	Direction of evidence/ Attribution:	This file is an image of a tusk and supports the charge. Attribution will be determined with further investigation
9	Policy Analysis:	Acceptable Use of Computing Resources: This file was used to transport data about tusks in secret in a possibly illegal way Personal Data Restrictions- Mark Jones has a personal file about tusks on company property Removable Media Prohibition— This file was found on a USB drive plugged into Dustin Green's computer.

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1	Artifact number:	4
2	Name of file:	car1.jpg
3	Hash:	3f7ebdb2ad808f1b042b6194686ed6f
4	File description: Provide the zone ID and URL	ZoneId=3 ReferrerUrl=https://www.google.com/ HostUrl=https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRvFsVjCjvTKuN2hJNoy8Xn8pO3mOVAf1ljlQ&s
5	Location/discovery method:	Located on Mark Jones USB drive, recovered via Autopsy.
6	Work done to recover:	Standard bit-stream copy. Extracted from Autopsy
7	Notice of suspicious work:	The photo/file had hidden in it another file. Seeming to avoid detecting and be a secret.
8	Direction of evidence/ Attribution:	This file is an image of a car that has a photo of tusk hidden in it and supports the charge. Attribution will be determined with further investigation
9	Policy Analysis:	Acceptable Use of Computing Resources: This file was used to transport data about tusks in secret in a possibly illegal way Personal Data Restrictions- Mark Jones has a personal file about cars on company property Removable Media Prohibition– This file was found on a USB drive plugged into Dustin Green’s computer.



1	Artifact number:	5
2	Name of file:	Car4.jpg
3	Hash:	c9e1a09188085fde3044e2e99b5b551f
4	File description: Provide the zone ID and URL	ZoneId=3 ReferrerUrl=https://www.google.com/ HostUrl=https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcTR8P58RFYoUEpdiXCB3XM_C_BGkDio_T2ppQ&s
5	Location/discovery method:	Located on Mark Jones USB drive, recovered via Autopsy. Stenography was used. Revealed using Hexed.it
6	Work done to recover:	Use Hexed.it and identified each photo by using magic numbers to identify the beginning and end of a photo

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7	Notice of suspicious work:	The photo/file was hidden in another file. Seeming to avoid detecting and be a secret.
8	Direction of evidence/ Attribution:	This file is an image of a tusk and supports the charge. Attribution will be determined with further investigation
9	Policy Analysis:	Acceptable Use of Computing Resources: This file was used to transport data about tusks in secret in a possibly illegal way Personal Data Restrictions- Mark Jones has a personal file about tusks on company property Removable Media Prohibition– This file was found on a USB drive plugged into Dustin Green’s computer.



1	Artifact number:	7
2	Name of file:	CAR5.jpg
3	Hash:	72ffb4918c90c02f3bb4c3c03bc98a75
4	File description: Provide the zone ID and URL	Zoneld=3 ReferrerUrl=https://www.google.com/ HostUrl=https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQGKvDfntCXuBsKTKEnI2RHorOTmicJWxon0Q&s
5	Location/discovery method:	Located on Mark Jones USB drive, recovered via Autopsy. Stenography was used. Revealed using Hexed.it
6	Work done to recover:	Use Hexed.it and identified each photo by using magic numbers to identify the beginning and end of a photo
7	Notice of suspicious work:	The photo/file was hidden in another file. Seeming to avoid detecting and be a secret.
8	Direction of evidence/ Attribution:	This file is an image of a tusk and supports the charge. Attribution will be determined with further investigation
9	Policy Analysis:	Acceptable Use of Computing Resources: This file was used to transport data about tusks in secret in a possibly illegal way Personal Data Restrictions- Mark Jones has a personal file about tusks on company property Removable Media Prohibition– This file was found on a USB drive plugged into Dustin Green’s computer.

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ARTIFACT 8



1	Artifact number:	8
2	Name of file:	CAR5.jpg
3	Hash:	a2c03406c8b49f9e9a81b74615092575
4	File description: Provide the zone ID and URL	ZoneId=3 ReferrerUrl=https://www.google.com/ HostUrl=https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQGKvDfntCXuBsKTKEnI2RHorOTmicJWxon0Q&s
5	Location/discovery method:	Located on Mark Jones USB drive, recovered via Autopsy.
6	Work done to recover:	Standard bit-stream copy. Extracted from Autopsy
7	Notice of suspicious work:	The photo/file had hidden in it another file. Seeming to avoid detecting and be a secret.
8	Direction of evidence/ Attribution:	This file is an image of a car that has a photo of tusk hidden in it and supports the charge. Attribution will be determined with further investigation
9	Policy Analysis:	Acceptable Use of Computing Resources: This file was used to transport data about tusks in secret in a possibly illegal way Personal Data Restrictions- Mark Jones has a personal file about cars on company property Removable Media Prohibition- This file was found on a USB drive plugged into Dustin Green's computer.

artifact 9



1	Artifact number:	9
2	Name of file:	car3.jpg
3	Hash:	bf847aaef34d00bd9546ba60e3194b3b
4	File description: Provide the zone ID and URL	ZoneId=3 ReferrerUrl=https://www.google.com/ HostUrl=https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRaqcCQU977qa7jT-IOgHS80tqwZcMS7ZsutA&s

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5	Location/discovery method:	Located on Mark Jones USB drive, recovered via Autopsy.
6	Work done to recover:	Standard bit-stream copy. Extracted from Autopsy
7	Notice of suspicious work:	The photo/file had hidden in it another file. Seeming to avoid detecting and be a secret.
8	Direction of evidence/ Attribution:	This file is an image of a car that has a photo of tusk hidden in it and supports the charge. Attribution will be determined with further investigation
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1	Artifact number:	11
2	Name of file:	car3.jpg
3	Hash:	583bcaddeb695e6d8f19aa4c0626362d
4	File description: Provide the zone ID and URL	Zoneld=3 ReferrerUrl=https://www.google.com/ HostUrl=https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRaqcCQU977qa7jT-IOgHS80tqwZcMS7ZsutA&s
5	Location/discovery method:	Located on Mark Jones USB drive, recovered via Autopsy. Stenography was used. Revealed using Hexed.it
6	Work done to recover:	Use Hexed.it and identified each photo by using magic numbers to identify the beginning and end of a photo
7	Notice of suspicious work:	The photo/file was hidden in another file. Seeming to avoid detecting and be a secret.
8	Direction of evidence/ Attribution:	This file is an image of a tusk and supports the charge. Attribution will be determined with further investigation
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1	Artifact number:	14
2	Name of file:	car4.jpg
3	Hash:	8a84cabaee410eb517cec9bf602d2791
4	File description: Provide the zone ID and URL	Zoneld=3 ReferrerUrl=https://www.google.com/ HostUrl=https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcTR8P58RFYoUEpdiXCB3XM_C_BGkDio_T2ppQ&s
5	Location/discovery method:	Located on Mark Jones USB drive, recovered via Autopsy. Stenography was used. Revealed using Hexed.it
6	Work done to recover:	Use Hexed.it and identified each photo by using magic numbers to identify the beginning and end of a photo
7	Notice of suspicious work:	The photo/file was hidden in another file. Seeming to avoid detecting and be a secret.
8	Direction of evidence/ Attribution:	This file is an image of a tusk and supports the charge. Attribution will be determined with further investigation
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1	Artifact number:	22
2	Name of file:	car4.jpg
3	Hash:	b1f391a9145b37eb59c23d0423028ebf
4	File description: Provide the zone ID and URL	Zoneld=3 ReferrerUrl=https://www.google.com/ HostUrl=https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcTR8P58RFYoUEpdiXCB3XM_C_BGkDio_T2ppQ&s

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7	Notice of suspicious work:	The photo/file was hidden in another file. Seeming to avoid detecting and be a secret.
8	Direction of evidence/ Attribution:	This file is an image of a tusk and supports the charge. Attribution will be determined with further investigation
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1	Artifact number:	32
2	Name of file:	Car3.jpg
3	Hash:	82b09e6246f11b94b5bb91a3fdf21e69
4	File description: Provide the zone ID and URL	Zoneld=3 ReferrerUrl=https://www.google.com/ HostUrl=https://encrypted-tbn0.gstatic.com/images?q=tbn:AND9GcRaqcCQU977qa7jT-IOgHS80tqwZcMS7ZsutA&s
5	Location/discovery method:	Located on Mark Jones USB drive, recovered via Autopsy. Stenography was used. Revealed using Hexed.it
6	Work done to recover:	Use Hexed.it and identified each photo by using magic numbers to identify the beginning and end of a photo
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6. Conclusion

Evidence Supporting Intellectual Property Theft

- **Artifact 1 – car1.jpg:** Document containing image of car
- **Artifact 2 – car2.jpg:** Document containing image of car
- **Artifact 3– car3.jpg:** Document containing image of tusk
- **Artifact 4 – car1.jpg:** Document containing image of car
- **Artifact 5 – car4.jpg:** Document containing image of car
- **Artifact 7 – CAR5.jpg:** Document containing image of tusk
- **Artifact 8 – CAR5.jpg:** Document containing image of car
- **Artifact 9 – car3.jpg:** Document containing image of car
- **Artifact 11–car3.jpg:** Document containing image of tusk
- **Artifact 14– car4.jpg:** Document containing image of tusk
- **Artifact 22– car4.jpg:** Document containing image of tusk
- **Artifact 32– car3.jpg:** Document containing image of tusk

Policy Breaches

- **Removable Media Prohibition:** The use of removable storage devices is strictly prohibited unless explicitly authorized by AT leadership. In approved cases, AT will issue a company-owned storage device to the employee for use.
- **Personal Data Restrictions:** Employees are prohibited from storing personal data on AT-owned computers.
- **Confidentiality of Company Property:** All confidential paper documents and electronic files belong exclusively to AT and must not be distributed outside the organization.

Taken together, these artifacts provide clear support for the illegal distribution of products unrelated to AT and document the associated policy violations committed by Mark Jones.

Steganography Question sheet

Notes for discussion/ lecture (see PowerPoint).

1. What are Zone Identifiers? What are the 4 Zones and what do they mean?- Zone identifiers are Windows only and acts as a marker showing where a file was downloaded from the internet or other potentially unsafe location. There are 5 zones. Zone 0, files originated on local computer. Zone 1 the file came from a local or network intranet. Zone 2 is the file came from a site explicitly marked as trusted by user or organization. Zone 3 is file was downloaded from public internet. Zone 4 the file came from a site explicitly marked as dangerous.

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2. What are magic numbers in documents?- the beginning hex of a file that tells the operating systems the file type for instead telling you if it is PDF or a JPEG. This ensures file integrity and detects file corruption

What is the file header for a jpg? - FF D8 FF

End of file marker in Jpg? - FF D9

What is Metadata?- It is data about data, information that describes, explain or give context to a file document or image helping the system understand how, when and who created or modified a file along with where sometimes

3. Question 2 - What is steganography? What are the two main types of steganography? Define each- The process of hiding messages, information, data or photos in another digital medium. This is done in an attempt to make a image or audio or video file look normal while hiding info. The two types or steganography are insertion and substitution. Substitution method hides information by replacing parts of the original files data with secret data. Insertion method hides information by adding new data to a file without removing or replacing existing data or content
4. What are the two main types of image compression? Define each- Lossless Compression is reducing file file size without removing any data. This is able to be decompressed and you can get a exact bit by bit copy of te original file. Lossy Compression is permanently removing some image data in a manner that a human eye is unable to see the change or removal of data
1. What is carving/salvaging in forensics? - Carving is the process of recovering files without relying on a file systems meta data. Salvaging is theprocess of recovering files using any available file system information.